Seungyong Moon

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Summary

My research develops reliable autonomous agents across perception, decision-making, and reasoning, with contributions to adversarial robustness, offline reinforcement learning, open-world generalization, and efficient reasoning.

Education

PhD **Seoul National University**, Computer Science and Engineering — Seoul, South Korea
- Advisor: Prof. Hyun Oh Song — Mar 2019 – Feb 2027

BA/BS **Seoul National University**, Economics/Mathematical Sciences — Seoul, South Korea
- Minor in Computer Science and Engineering — Mar 2011 – Feb 2019
- Summa Cum Laude (ranked 1st out of 34 students)

Experience

Qualcomm AI Research, Research Intern — Amsterdam, Netherlands
Sept 2024 – Jan 2025
5 months
- Developed an RL framework that trains LLMs to adaptively execute classical algorithms via a Python REPL based on input complexity, reducing execution steps and computational overhead while maintaining correctness.
- Built a multi-turn RL fine-tuning pipeline for LLMs from scratch, enabling complex multi-step reasoning and tool use, before such support existed in mainstream libraries.

KRAFTON AI, Research Intern — Seoul, South Korea
June 2023 – Sept 2023
4 months
- Developed a self-supervised RL algorithm for discovering hierarchical skills in a Minecraft-like game, achieving SOTA with 4% of the parameters of the previous best.
- Designed an agentic framework that improves the spatial reasoning of GPT-4 in grid-based game environments, leveraging self-reflection on interaction history to infer environment dynamics.

DeepMetrics, Research Intern — Seoul, South Korea
June 2022 – Sept 2022
4 months
- Built a data preprocessing pipeline for 20K+ hours of ventilator waveform data, enabling large-scale training of ventilator control policies.
- Developed imitation learning, reinforcement learning, and off-policy evaluation methods for autonomous ventilator control.

NAVER Search & Clova, Research Intern — Gyeonggi-do, South Korea
July 2018 – Aug 2018
2 months
- Developed a data augmentation method for paraphrase identification in seq2seq models, improving accuracy by 2%.

Publications

Learning to Better Search with Language Models via Guided Reinforced Self-Training — NeurIPS 2025
Seungyong Moon, Bumsoo Park, Hyun Oh Song
arxiv.org/abs/2410.02992

Improving the Efficiency of Algorithmic Reasoning in Language Models via Reinforcement Learning — Preprint
Seungyong Moon, Michaël Defferrard, Corrado Rainone, Roland Memisevic

Discovering Hierarchical Achievements in Reinforcement Learning via Contrastive Learning NeurIPS 2023
Seungyong Moon, Junyoung Yeom, Bumsoo Park, Hyun Oh Song
arxiv.org/abs/2307.03486

Rethinking Value Function Learning for Generalization in Reinforcement Learning NeurIPS 2022
Seungyong Moon, JunYeong Lee, Hyun Oh Song
arxiv.org/abs/2210.09960

Query-Efficient and Scalable Black-Box Adversarial Attacks on Discrete Sequential Data via Bayesian Optimization ICML 2022
Deokjae Lee, *Seungyong Moon*, Junhyeok Lee, Hyun Oh Song
arxiv.org/abs/2206.08575

Preemptive Image Robustification for Protecting Users against Man-in-the-Middle Adversarial Attacks AAAI 2022
*Seungyong Moon**, Gaon An*, Hyun Oh Song
arxiv.org/abs/2112.05634

Uncertainty-Based Offline Reinforcement Learning with Diversified Q-Ensemble NeurIPS 2021
Gaon An*, *Seungyong Moon**, Jang-Hyun Kim, Hyun Oh Song
arxiv.org/abs/2110.01548

Parsimonious Black-Box Adversarial Attacks via Efficient Combinatorial Optimization ICML 2019
*Seungyong Moon**, Gaon An*, Hyun Oh Song
arxiv.org/abs/1905.06635

Awards and Scholarships

- NeurIPS Top Reviewers (2022, 2025)
- NeurIPS Scholar Award (2023)
- NAVER PhD Fellowship Award (2022)
- Youlchon AI Star Scholarship (2022)
- Qualcomm Innovation Fellowship Korea Finalists (2020, 2022)
- KFAS Computer Science Graduate Student Scholarship (2019 – 2024)

Skills

Languages: Python, C++, MATLAB

ML Frameworks: PyTorch, JAX, TensorFlow

Infrastructure: Docker, Slurm

Patents

1. Apparatus and method with out-of-distribution data detection (US Patent Application 18/360,229, KR Patent Application 10-2023-0004866)
2. Image recognition method and apparatus, image preprocessing apparatus, and method of training neural network (US Patent 11,921,818, KR Patent 10-2891575)
3. Method and apparatus for classifying image (KR Patent 10-2505303)

Academic Services

- Conference Reviewer: NeurIPS (2021 – 2026), ICML (2022 – 2026), ICLR (2024 – 2026), AAAI (2022, 2024), AAAI-AIA (2026), AAAI-Demo (2026), AISTATS (2025 – 2026), RLC (2024)
- Journal Reviewer: Neurocomputing (2021), Machine Learning (2023), IEEE Transactions on Intelligent Vehicles (2023), Pattern Recognition (2026), Transactions on Machine Learning Research (2026)